

Data Driven Economics

Text Mining in Economics

Lecture 3

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¹The opinions expressed are those of the author and do not reflect the views of the Bank of Italy, the ECB or the Eurosystem.

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Roadmap

1. **Techniques** for **Text Mining** in general.
2. **Mainly** unsupervised
3. Dimensionality reduction
4. Conclusion

Text Mining for Central Banks

Bholat, D., Hansen, S., Santos, P., Schonhardt-Bailey, C., 2015, "Text Mining for Central Banks", CCBS Handbook No. 33, Bank of England.

Dimensionality reduction techniques

Steps:

1. Analytical Pre-Processing ✓
2. Boolean Techniques ⇐
3. Dictionary Techniques ⇐
4. Weighting Words ⇐
5. Vector Space Models ⇐
6. Latent Semantic Analysis (LSA) ⇐
7. Latent Dirichlet Allocation (LDA) ⇐
8. Descending Hierarchical Classification
9. Supervised Machine Learning

Text as Data for Central Bank Research

Deduction vs abduction

- ▶ **Boolean** and **dictionary** approaches are deductive techniques because they start with a predefined list of words (i.e. a **dictionary!**), motivated by a general theory as to why those words matter.
 - ▶ **Strength #1: simplicity and scalability.**
 - ▶ **Strength #2: Few lines of code**
 - ▶ **Weakness #1: focus only on words pre-judged to be informative but ignoring all the other words**
- ▶ **Latent Semantic Analysis, Latent Dirichlet Association** and **Descending Hierarchical Classification** infer thematic patterns in a particular corpus without claiming that these patterns hold in other documents.
 - ▶ **Strength #1: analyze all words within the sampling frame**
 - ▶ **Strength #2: yield more sophisticated statistical outputs**
 - ▶ **Weakness #3: programming complexity**

Pre-Processing Text: Example from Hansen et al. (2018)

- ▶ Hansen et al. “Transparency and Deliberation within the FOMC: A Computational Linguistic Approach, *Quarterly Journal of Economics* (2018) analyze FOMC members’ behavior in FOMC monetary policy minutes.
- ▶ Natural experiment: in 1993 FOMC minutes were made public and FOMC members did not know that before.

TABLE I
INFORMATION MADE AVAILABLE BY DIFFERENT CENTRAL BANKS AS OF 2014

	Federal Reserve	Bank of England	European Central Bank
Release minutes?	Yes	Yes	No
Release transcripts?	Yes	No	No

Pre-Processing Text: Example from Hansen et al. (2018)

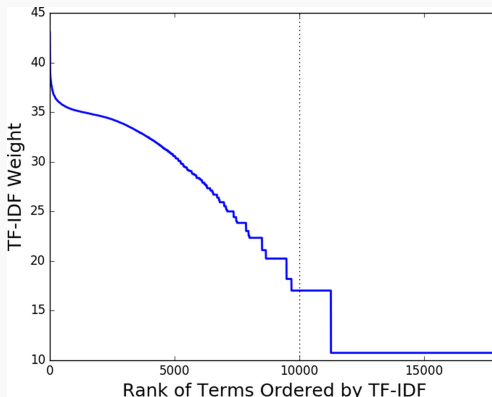


FIGURE I

Ranking of Stems with Term Frequency-Inverse Document Frequency

Let n_v be the count of term v in the data set. The term frequency tf_v is $1 + \log(n_v)$. The document frequency is $df_v = \log\left(\frac{D}{D_v}\right)$ where D is the number of documents and D_v is the number of documents in which term v appears. The tf-idf weight of term v is $tf_v \times df_v$. This figure plots the tf-idf weight of each stem in FOMC1 and FOMC2, and the dotted line indicates that the threshold we choose to drop stems from the data for the analysis.

Pre-Processing Text: Example from Hansen et al. (2018)

- ▶ Effects of pre-processing text on text corpus.
- ▶ Example of collocation: 'labor market' is considered as a unique word

TABLE II
DATA DIMENSIONALITY REDUCTION OF EACH PREPROCESSING STEP

	Raw text	Identify collocations	Remove stopwords	Stemming	TF-IDF adjustment
Total words	3,917,629	3,814,074	1,732,323	1,732,323	1,672,869
Unique words	24,314	25,019	24,822	15,394	8,206

Notes. Our raw text contains 3,917,629 words, 24,314 of which are unique. This table shows how these numbers evolve through preprocessing. The stopword list we use is from <http://snowball.tartarus.org/algorithms/english/stop.txt>. The stemming algorithm is the Porter stemmer implemented in Python's Natural Language Toolkit (Bird, Klein, and Loper 2009). TF-IDF weighting is as described in the main text.

Text mining techniques to reduce dimensionality

1. Boolean techniques
2. Dictionary techniques
3. Weighting words
4. Vector space models
5. Latent Semantic Analysis (LSA)
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8. Supervised machine learning

Boolean Techniques

- ▶ By far the simplest method for mining texts.
- ▶ You perform a Boolean search for one or more key terms.
- ▶ Boolean search techniques combine individual terms or phrases with logical operators like AND, OR, and NOT to form search expressions.
- ▶ Each document is then assigned a number 1 or 0 depending on whether the expression is true ($= 1$) or not ($= 0$).
- ▶ Boolean search is incorporated in most major internet search engines. Therefore one advantage is that the researchers do not have to have access to the raw texts if they have already been indexed for them by the search engine algorithms.
- ▶ For example Google or Bing will return an estimate of the number of websites that satisfy Boolean search criteria.

Boolean Techniques: A typical example

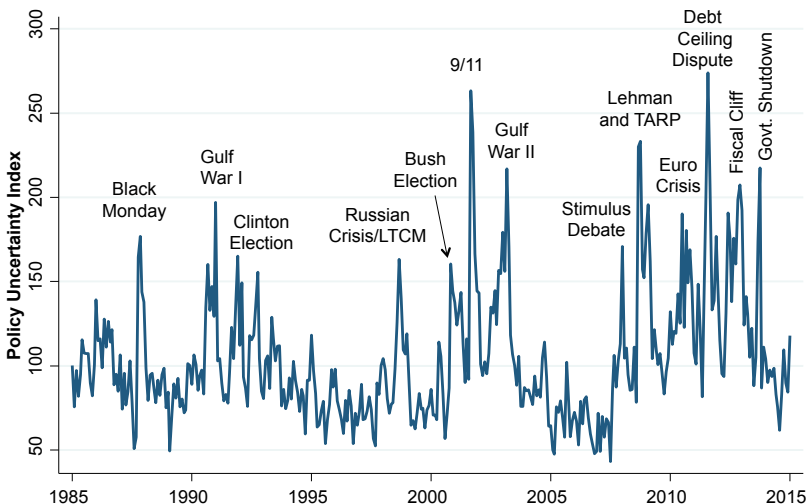
- ▶ A typical example of research relevant for central banks and using Boolean techniques is the work by Baker, Bloom, and Davis (2016) who estimate an indicator of Economic Policy Uncertainty (EPU) by analyzing the major US and European newspapers.
- ▶ For each paper on each day beginning in 1985, they count the number of articles containing words related to Uncertainty, Economics and Policy. t

Boolean Techniques: A typical example

- ▶ They compute the number of articles per day (or month) that satisfy the following search criteria:
 - (E) Articles containing 'economic', OR 'economy' AND
 - (P) Articles containing 'congress', OR 'deficit', OR 'federal reserve', OR 'legislation', OR 'regulation', OR 'white house' AND
 - (U) Articles containing 'uncertain', OR 'uncertainty'
- ▶ The time series of the **normalized** counts – scaled by the total number of articles in the newspaper each day (or month) – then represents a proxy for Economic Policy Uncertainty.
- ▶ The EPU is able to capture important political and financial market events.

Economic Policy Uncertainty (EPU) index

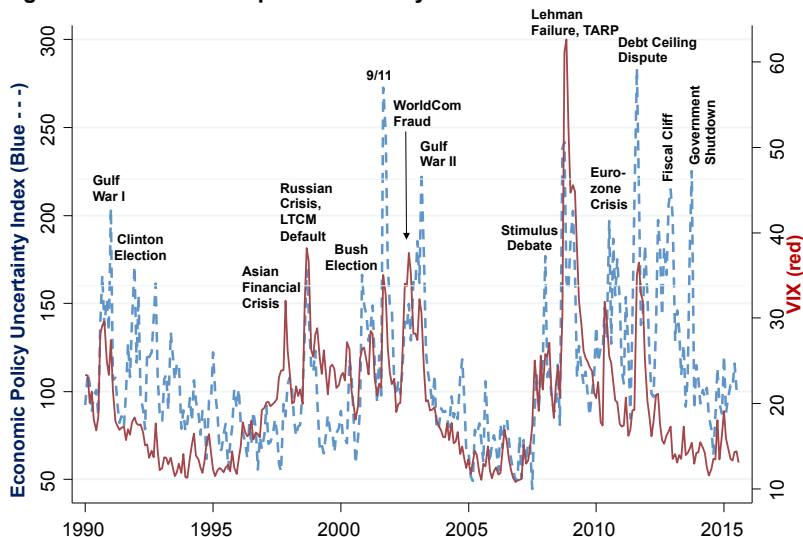
Figure 1: Economic Policy Uncertainty Index for the US



Notes: Index reflects scaled monthly counts of articles containing 'uncertain' or 'uncertainty', 'economic' or 'economy', and one or more policy relevant terms: 'regulation', 'federal reserve', 'deficit', 'congress', 'legislation', or 'white house'. The series is normalized to mean 100 from 1985-2009 and based on queries run on 2 February, 2015 for the USA Today, Miami Herald, Chicago Tribune, Washington Post, LA Times, Boston Globe, SF Chronicle, Dallas Morning News, NY Times, and the Wall Street Journal.

Economic Policy Uncertainty (EPU) index

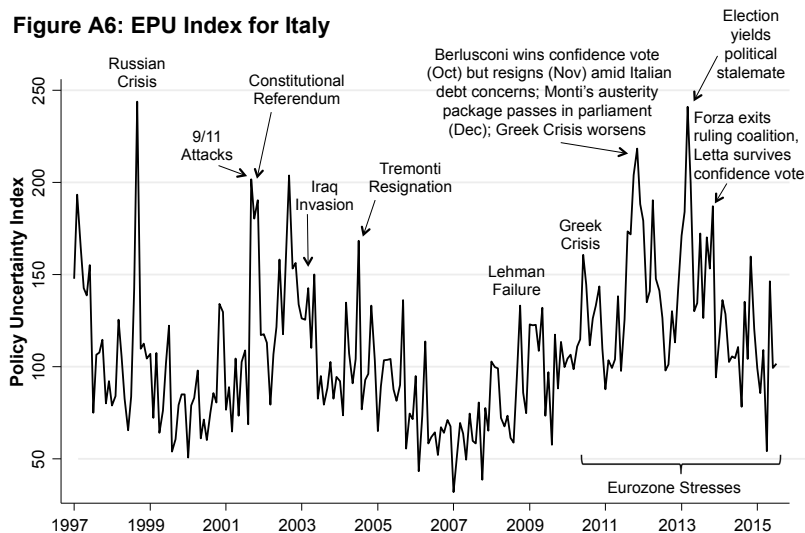
Figure 6: U.S. EPU Compared to 30-Day VIX



Notes: The figure shows the U.S. EPU Index from Figure 1 and the monthly average of daily values for the 30-day VIX.

Economic Policy Uncertainty (EPU) index

Figure A6: EPU Index for Italy



Notes: Index reflects scaled monthly counts of articles containing 'uncertain' or 'uncertainty', 'economic' or 'economy', and one or more policy-relevant terms: 'tax', 'policy', 'regulation', 'spending', 'deficit', 'budget', or 'central bank'. The series is normalized to mean 100 from 1997 to 2009 and based on the following newspapers: La Stampa and Corriere Della Sera.

Other examples of Boolean Techniques

- ▶ Another example of Boolean text mining is the work by BoE during the run-up to the Scottish independence vote.
- ▶ At that time there were concerns about the negative consequences for the financial system and for the banks of a 'yes' result.
- ▶ For example there was fear of runs on Scottish banks (Bank of England, 2014).
- ▶ BoE monitored the situation using Twitter traffic.
- ▶ Tweets containing terms and combinations of terms suggestive of bank runs were monitored for a full week and in particular in the night of the vote. In the end there was little traffic but the BoE found a big false positive!
- ▶ Bank of England was looking for: **"RBS"** i.e. **"Royal Bank of Scotland"** and runs and they found **"RBs"** i.e. **"Running Backs"**!

Possible Pitfalls in Text Analysis?

BoE looking for RBS on Twitter and finding RBs!

Trying to predict a bank run, the Bank of England discovers the Minnesota Vikings



Minnesota Vikings running back Matt Asiata runs over New England Patriots cornerback Kyle Arrington during the second quarter at TCF Bank Stadium. Tweets about Vikings running backs got caught up in an analysis project by the Bank of England.

Boolean Techniques: Pros and Cons

- ▶ Boolean text-based measures of risk and uncertainty have four comparative strengths over traditional measures of uncertainty like the VIX.
- ▶ First, by modifying search terms one can capture broader concepts of uncertainty beyond those specific to financial markets.
- ▶ Second, the source of uncertainty is often stated explicitly in text. For example, 'uncertainty caused by the referendum vote'. By contrast, with measures of uncertainty based on financial asset prices, the source of uncertainty is not always clear.
- ▶ Third, by extending media text searches back in time. We can obtain a longer time series than the VIX which is available only from late 1980's. Baker et al.'s EPU goes back to 1900.
- ▶ Fourth, we can obtain cross-country measures of uncertainty by focusing on separate countries' media texts. By contrast, VIX is only a proxy for uncertainty in the US equity prices and it might be a poor proxy of uncertainty for other countries.

Text mining techniques to reduce dimensionality

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Dictionary Techniques

- ▶ Dictionary techniques typically proceed in two steps:
 1. Start by defining a list (a dictionary) of key words to capture the content of interest;
 2. Then represent each document in terms of the (normalized) frequency of words in the dictionary.
- ▶ Unlike Boolean techniques, dictionary ones measure the **intensity of word use** which might be a better measure of the corpus' content.
- ▶ Nevertheless, unlike Boolean techniques, dictionary ones require to have access to the raw texts.

Dictionary Techniques

- For example: imagine you want to analyze the labor market and you set your dictionary to:

$$D = \{\text{labor, wage, unemployment}\}$$

We can then represent each document d as the share s_d of words that are in the dictionary:

$$s_d = \frac{(\# \text{ of 'labor' occurrences} + \# \text{ of 'wage' occ.} + \# \text{ of 'unemployment' occ.})}{\text{total words in document } d}$$

Dictionary Techniques: examples

- ▶ In the literature there are any examples in which dictionary techniques have been used to analyze financial and economic texts.
- ▶ For example, Tetlock (2007) wrote a highly-cited paper that uses dictionaries (*aka* lexicons) to gauge the tone (i.e. the sentiment) of the *Wall Street Journal* column “Abreast of the Market”.
- ▶ Tetlock (2007) employs the Harvard IV-4 dictionaries containing word lists reflecting many classes, including positive and negative sentiment, pain and pleasure, and rituals and natural processes, among others.
- ▶ Tetlock (2007) counts the number of words in each day’s column from 1984 to 1999.
- ▶ He finds that most of the variation in word counts across columns reflects swings from **optimism** to **pessimism**.
- ▶ He also finds that the impact of an increase in negative news moves the next day’s returns in a statistically significant way.

Dictionary Techniques: examples

- ▶ After Tetlock (2007), many papers have used the same strategy of counting words of interest in financial market and correlating them to asset prices.
- ▶ Nevertheless, as we already noticed many times, there are many instances in which raw word counts may be misleading.
- ▶ If we consider **sentiment analysis**, also called **opinion mining** or subjectivity analysis, the goal is to detect the feeling expressed about some target.
- ▶ The target might be a person, an event, an institution or an object.
- ▶ The simplest type of sentiment analysis is **polarity detection** which involves binary classification of positive and negative sentiments.

Dictionary Techniques: examples

- ▶ If you want to accurately gauge sentiment, the task might be complicated by a number of linguistic subtleties such as: (i) negation, (ii) irony, (iii) ambiguity, (iv) idioms, and (v) neologisms.
- ▶ For example, consider the following paragraph:

One would have expected building society X to have been an excellent institution. It had a top-notch CEO and world-renowned analysts. Its services were highly rated by retail clients and its operations were efficient. Yet it was a total failure.

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- ▶ Notice that just counting the balance of +ve and -ve words does NOT convey the real sentiment of the passage which is fundamentally negative.

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Weighting words

- ▶ Simple counting approaches may be inappropriate because they can overstate the importance of a small number of very frequent words.
- ▶ This is troublesome for two reasons:
 - ▶ **Zipf's law**, shorthand for the empirical observation that the frequency of a word is inversely proportional to its relative rank in a corpus. Thus a small difference in the relative rank of a word can translate into a big difference in terms of the actual word count. So relying on raw word counts may overstate their comparative importance.
 - ▶ If a word is used in many documents of a corpus, then its power to discriminate one document from another one is less than if it only appears in a handful of documents.

Weighting words

- ▶ To address these concerns a **common weighting scheme** is used in the text mining literature: **term frequency-inverse document frequency** (tf.idf) given by

$$td.idf(t, d) = \underbrace{[1 + \log f(t, d)]}_{\text{less weight to more frequent words}} \cdot \underbrace{\log \left(\frac{D}{df_t} \right)}_{\uparrow \text{weight of less frequent word}}$$

where D is the total # of documents in the corpus, df_t is the number of documents in which the term t appears and $f(t, d)$ is the frequency of term t in document d .

Weighting words

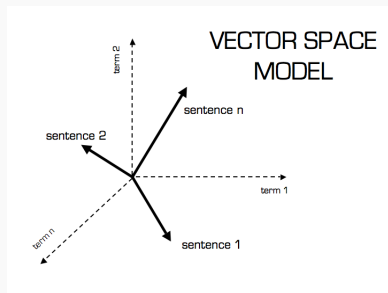
- ▶ A very famous example of *tf.idf* weighting is the Loughran and McDonald (2011) dictionary.
- ▶ Loughran and McDonald (2011) start criticizing the Harvard IV-4 dictionaries used by Tetlock (2007).
- ▶ In fact, Tetlock's (2007) dictionary contains words like 'tax', 'cost', or 'liability' that, though they convey a negative sentiment in a general context, they are more neutral in tone in the **context** of financial markets where they describe day-to-day accounting practices.
- ▶ Loughran and McDonald (2011) therefore suggest a finance-specific dictionary and show that they can better predict asset returns than generic dictionaries.
- ▶ However, after *tf.idf* weighting, the generic dictionaries' performance improves substantially.
- ▶ The reason is simple. Words like 'tax' appear in many documents and thus they get a lower weight than other truly negative words.

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Vector Space Models

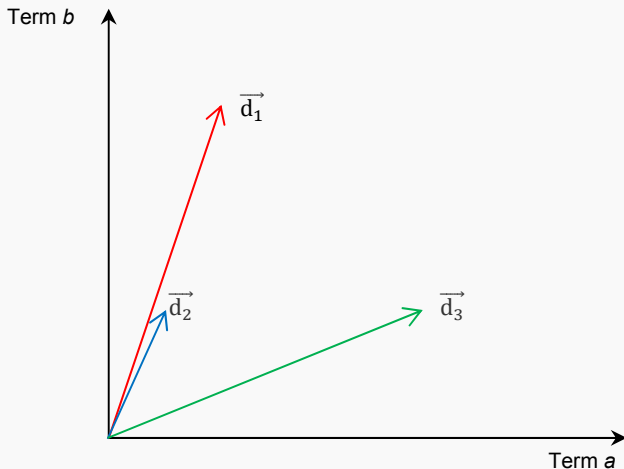
- ▶ The representation of a set of documents as vectors in a common vector space is known as the vector space model (or term vector model).
- ▶ Consider the vector derived from a document with one component in the vector for each term.
- ▶ The set of documents in the collection may then be viewed as a set of vectors in a vector space, in which there is one axis for each term.
- ▶ This representation loses the relative ordering of the terms in each document. This type of feature vectors are the basis for the bag of words models.



Vector Space Models

- ▶ So far, we have seen techniques to identify the key topics **within** texts using a pre-defined set of keywords.
- ▶ With the Vector Space model we consider techniques for measuring the similarity of topics **between** texts.
- ▶ One way to measure document **similarity** is to use simple Euclidean distance to find clusters of documents.
- ▶ However even this distance has some limitations.
- ▶ For example, suppose that the corpus consists of 3 documents each containing 2 terms a and b . Doc. #1 and #2 use terms a and b in nearly the same proportions but #1 is longer and their distance is quite significant. Doc. #3 which uses term a relative to term b more than doc. #2 would be more similar to doc. #1 just because of the length.

Vector Space Models



Vector Space Models

- ▶ This example shows the limitation and distortions arising from the Euclidian distance.
- ▶ A measure that avoids these kinds of problems is **cosine similarity** (CS), which captures the angle formed by two vectors.
- ▶ Looking at the previous figure, we can see that the angle between doc. #1 and #2 is very small. The two vectors point towards the same direction because they use the two terms in almost identical proportions.
- ▶ Since the term frequencies differ between doc.#3 and #1 the angle is larger.
- ▶ At one extreme if one document vector contains only term a and another one only term b , the vectors would be orthogonal.
- ▶ Therefore, measuring the **cosine of the angle** θ formed by two documents in the vector space provides a **similarity measure** that is independent of document length.

Cosine Similarity

- ▶ The formula for computing **Cosine Similarity** is

$$\cos \theta = \frac{\vec{d}_1 \cdot \vec{d}_2}{\|\vec{d}_1\| \|\vec{d}_2\|}$$

where $\cdot$ is the product operator and $\|\vec{d}_i\|$ is the length of the vector representing document i .

- ▶ An example of vector space model in economics is Hoberg and Philips (2010). They take company product statements from 10-K filings with the US Securities and Exchange Commission and compute the CS.
- ▶ They use the similarity score as a proxy for the companies' sector. They argue that their measure gives better results than traditional industry codes.

Cosine Similarity is no panacea

⇒ **Synonymy**

⇒ **Polysemy**

Table 6 This table shows two documents with similar content but low cosine similarity due to synonymy

(a) Document 1				
school	university	college	teacher	professor
0	5	5	1	2
(b) Document 2				
school	university	college	teacher	professor
10	0	0	4	0

- ▶ 2 documents talk about same topic using different words ⇒ low CS!

Table 7 This table shows two documents with dissimilar content but relatively high cosine similarity due to polysemy

(a) Document 1					
tank	marine	frog	animal	navy	war
5	5	3	2	0	0
(b) Document 2					
tank	marine	frog	animal	navy	war
5	5	0	0	4	3

- ▶ The same word can have multiple meanings in different contexts. (animals vs war)
- ▶ Unrelated documents with high CS!

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Topic models

- ▶ All topic models are based on the same basic assumptions:
 - ▶ each document consists of a mixture of topics
 - ▶ each topic consists of a collection of words
- ▶ In other words, topic models are built around the idea that the semantics of our document are actually being governed by some hidden, or “latent” variables that we are not observing.
- ▶ As a results, the goal of topic modeling is to uncover these latent variables – **topics** – that shape the meaning of our document and corpus.

Latent Semantic Analysis

- ▶ One assumption of vector space models is that words in texts are independent of each other.
- ▶ However 'navy' and 'marine' describe a latent topic such as 'war'.
- ▶ Latent variable models assume that **words are not independent** but they are **linked** together by underlying and unobserved topics.
- ▶ Virtues:
 - ▶ They deal with synonymy by associating each word in the vocabulary to any given latent topic.
 - ▶ They capture polysemy by allowing each word to have associations with multiple topics.
 - ▶ They associate each document to topics rather than words.
 - ▶ They allow algorithms to find the best association between words and latent variables without using pre-defined dictionaries (word lists) or classes.

Latent Semantic Analysis

- ▶ Latent Semantic Analysis (LSA) is an early example of latent variable approach (Deerwester et al., 1990).
- ▶ Starting from the term-document matrix it applies SVD to find the principal components (kind of factors) of the rows and columns.
- ▶ Thus LSA computes the linear combinations of terms that explain most variance of terms across documents, as well as linear combinations of documents that explain most of the variance of documents across terms.
- ▶ Idea: approximate the term-document matrix using the PCs and to measure document similarity with the approximation rather than the true term-document matrix.
- ▶ The hypothesis is that PCs represent shared topics and the discarded components represent idiosyncratic words choices.

LSA examples

- ▶ One recent paper on LSA is Acosta (2014) who studies the effect of greater transparency on US Federal Reserve Open Market Committee (FOMC) meetings.
- ▶ Acosta (2014) studies if greater awareness by FOMC members that their comments in meetings were being recorded and would be publicly disclosed changed their behavior.
- ▶ Acosta (2014) applies LSA and uses the top 200 components to measure document similarity.
- ▶ He finds an increased conformity after the publication of the transcripts.

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Latent Dirichlet Allocation (LDA)

- ▶ One great weakness of LSA is that the topics it produces are **not probabilistic**.
- ▶ LDA rectifies this
- ▶ LDA is a mixed-membership model in which words and documents are assigned probabilities and related to multiple topics.
- ▶ This contrasts with deterministic single-membership models where words and documents are assigned only to one topic.
- ▶ This **probabilistic** aspect of LDA is important.

Latent Dirichlet Allocation (LDA)

- ▶ Suppose one has two topics *'inflation'* and *'unemployment'*. If we consider the word *'rate'* it is unclear into which topic *'rate'* should go. We might have either *'inflation rate'* or labour *'participation rate'*.
- ▶ Allowing this probabilistic assignment of words to topics gives semantic flexibility.
- ▶ Formally, each document has its own probability distribution over topics.
- ▶ Then, for each word in each document, a topic assignment is made, and then, conditional on the assignment, a word from the corresponding topic.

Latent Dirichlet Allocation (LDA)

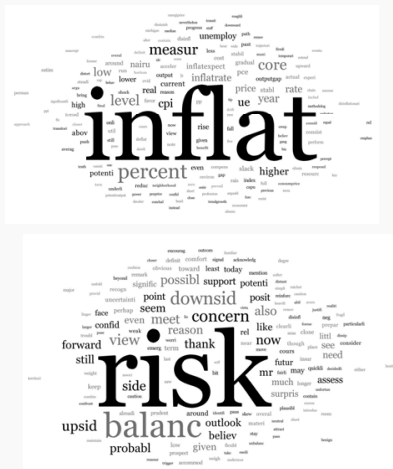
- ▶ Example: there are two speakers A and B both talking about topics 1 and 2.
- ▶ Speaker A spends $\frac{2}{3}$ of her time on topic 1.
- ▶ Speaker B spends $\frac{2}{3}$ of her time on topic 2.
- ▶ Thus, A's observed words can be conceived as being chosen as follows:
 - ▶ For each word, A draws a topic, and, with probability 0.67, this is topic 1, and with probability 0.33, this is topic 2.
 - ▶ Once a topic is drawn for a word, the word itself is drawn from the probability distribution associated with each topic.
 - ▶ For speaker B, the probability that a word is about topic 1 is 0.33 and about topic 2 is 0.67.
 - ▶ Once a word is assigned to a given topic, it is drawn from that topic's distribution which is common across the speakers.

Latent Dirichlet Allocation: from Hansen et al. (2016)

- ▶ LDA has been widely applied in computer science, but not in economics.
- ▶ Hansen et al. (2016) study the effects of increased central bank transparency.
- ▶ Like Acosta (2014), they find evidence of increased conformity after the release of the FOMC transcripts.
- ▶ In the next slide there are two topics they estimated. Topics are represented as word clouds in which the size of the word is proportional to its probability in the topic.
- ▶ No classes were imposed during estimation. The ones in the figure are the 25th and 40th word clusters estimated.

Latent Dirichlet Allocation: from Hansen et al. (2016)

Figure 9 This figure displays two topics estimated from the corpus of FOMC transcripts during the Alan Greenspan era



Source: Hansen, McMahon and Prat (2014).

Latent Dirichlet Allocation: from Hansen et al. (2016)

- ▶ What are the topics about?

Latent Dirichlet Allocation: from Hansen et al. (2016)

- ▶ What are the topics about?
- ▶ Of course, topic 25 (top) is about inflation, topic 40 (bottom) is about risk.
- ▶ Clearly, we can use the output from LDA to measure underlying content in a directly meaningful way.

Latent Dirichlet Allocation: Blei (2003)

Topics

gene 0.04
dna 0.02
genetic 0.01
...

life 0.02
evolve 0.01
organism 0.01
...

brain 0.04
neuron 0.02
nerve 0.01
...

data 0.02
number 0.02
computer 0.01
...

Documents

Seeking Life's Bare (Genetic) Necessities

COLD SPRING HARBOR, NEW YORK—How many **genes** does an **organism** need to **survive**? Last week at the genome meeting here, two genome researchers with radically different approaches presented complementary views of the basic genes needed for **life**. One research team, using **computer** analyses to compare known **genomes**, concluded that today's **organisms** can be sustained with just 250 genes, and that the earliest life forms required a mere 128 **genes**. The other researcher mapped genes in a simple parasite and estimated that for this organism, 800 genes are plenty to do the job—but that anything short of 100 wouldn't be enough. Although the numbers don't match precisely, those **predictions**

* Genome Mapping and Sequencing, Cold Spring Harbor, New York, May 8 to 12.

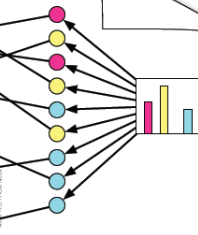
SCIENCE • VOL. 272 • 24 MAY 1996

"are not all that far apart," especially in comparison to the 75,000 **genes** in the human genome, notes Siv Andersson, a University of Sweden researcher. "But coming up with a consensus answer may be more than just a **genetic** numbers game, particularly as more and more **genomes** are completely mapped and sequenced. "It may be a way of organizing any newly **sequenced genome**," explains Arcady Mushegian, a **computational** molecular biologist at the National Center for Biotechnology Information (NCBI) in Bethesda, Maryland. Comparing an



Stripping down. **Computer analysis** yields an estimate of the minimum modern and ancient genomes.

Topic proportions and assignments



From Thorsrud (JBES, 2018)

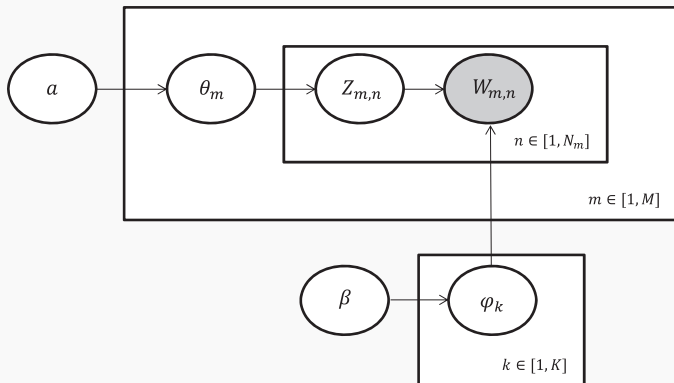


Figure 1. The LDA model visualized using plate notation.

Latent Dirichlet Allocation

- ▶ In the Figure on slide 49 the outer box or plate represents the whole corpus as M distinct documents (articles, tweets, etc.).
- ▶ $N = \sum_{m=1}^M N_m$ is the total number of words in all documents
- ▶ K is the total number of latent topics
- ▶ θ_m is the distribution of topics
- ▶ φ_k is the distribution of words for each topic
- ▶ Both θ_m and φ_k are assumed to have conjugate Dirichlet distributions with hyper parameters α and β , respectively.
- ▶ Each document consists of a repeated choice of topics $Z_{m,n}$ and words $W_{m,n}$ drawn from the multinomial distribution using θ_m and φ_k
- ▶ The circle around $W_{m,n}$ indicates the only observable variables in the model

Latent Dirichlet Allocation

- More formally, the joint distribution of all known and hidden variables given the hyper parameters is

$$\begin{aligned}
 P(W_m, Z_m, \Phi; \alpha, \beta) = & \\
 & \text{document plate (1 document)} \\
 = & \underbrace{\prod_{n=1}^{N_m} P(W_{m,n} | \varphi_{z_{m,n}}) P(Z_{m,n} | \theta_m) \cdot P(\theta_m; \alpha)}_{\text{word plate}} \cdot \underbrace{P(\Phi; \beta)}_{\text{topic plate}} \\
 & (1)
 \end{aligned}$$

where $\Phi = \{\varphi\}_{k=1}^K$ is a $K \times V$ matrix, and V is the size of the vocabulary

Latent Dirichlet Allocation: algorithms

- ▶ Different algorithms exist for solving LDA model.
- ▶ For example Griffiths and Steyvers (2004) do not treat θ_{mjh} and φ_k as parameters to be estimated but they integrate them out of eq. (1)
- ▶ Considering the whole corpus, this results in an expression for

$$P(V, Z; \alpha\beta) = P(Z|W; \alpha\beta)P(W; \alpha, \beta)$$

- ▶ This can be solved using Gibbs simulations.
- ▶ Estimates of θ_m and φ_k can be obtained from the posterior distribution

Text mining techniques to reduce dimensionality

1. Boolean techniques
2. Dictionary techniques
3. Weighting words
4. Vector space models
5. Latent Semantic Analysis (LSA)
6. Latent Dirichlet Allocation (LDA)
7. Descending Hierarchical Classification
8. Supervised machine learning

Descending Hierarchical Classification

- ▶ Beyond LSA and LDA, there are other Text Mining techniques that use the assumption that words are not independent of each other but reflect underlying topics or themes.
- ▶ For example, Schonhardt-Bailey (2013) and Vallès and Schonhardt-Bailey (2015) study central banks's texts using Alceste software.
- ▶ Alceste software tries to discover stable classes of terms that are maximally associated within, while being minimally associated with other classes.

Descending Hierarchical Classification

- ▶ To do this Alceste constructs a matrix that crosses all statements – so called Elementary Context Units (ECUs) – and words, where the cells indicate the presence or absence of that word in the statement, like a term-document ($td(t, d)$) matrix, but at a smaller unit.
- ▶ The cells indicate absence (0) or presence (1) of that word in the statement.

Table 8 Word by ECU matrix

ECU	1	2	3	4	5	...	j
word							
1	0	1	1	1	1	...	0
2	1	1	0	0	0	...	1
...	...	...	...	...	...	...	...
i	0	1	1	0	1	...	1
Totals	2	31	5	10	67	...	2

Descending Hierarchical Classification

- ▶ Then Alceste partitions the contents of this table into two classes.
- ▶ The goal is to maximize the similarity of ECUs in the same class while at the same time maximizing the difference between classes.
- ▶ Then the algorithm minimizes the number of overlapping words using a chi-squared to compare the expected and observed distribution.
- ▶ Then the algorithm maximizes the chi-square values repeating the partitioning process (descending hierarchical classification), that is by testing if splitting classes into more granular subclasses improves the χ^2 values.
- ▶ The iterative DHC stops when a predetermined number of iterations no longer results in statistically significant divisions.

Conclusion and way further

Key messages to take home:

- ▶ We saw how to pre-process text to feed it to text mining algorithms
- ▶ We also saw some popular techniques to extract information from text
- ▶ Most of them are dimensionality reduction techniques.
- ▶ In the next lectures we will see applications of Text Mining in Economics.
- ▶ Be aware that some of them will be replicated in the final project.

Some References I



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